

Amartya Tashi Mitra

Curriculum Vitae

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Areas of Interest

Evolution, Development, Morphology, Vision, Ecology

Education

1. 2022 - Current, **University of Cincinnati, USA**
PhD in Biology supervised by Professor Elke K. Buschbeck
Thesis - Towards understanding the development and function of insect camera-type eyes
2. June - July 2025, **Marine Biological Laboratory, USA**
Embryology: Concepts and Techniques in Modern Developmental Biology
3. 2019 - 20, **University College London, UK**
MRes in Biodiversity Evolution and Conservation. **Overall mark: Distinction (75%)**
 - Masters research project at the **Natural History Museum, London**, supervised by Dr Kate Thomas, Dr David Gower and Dr Jeffrey Streicher on "Lens morphology is influenced by ecology in frogs and toads (Amphibia: Anura)", **Distinction (80%)**
 - Masters research project at the **Zoological Society of London**, supervised by Dr Gonçalo Rosa, Professor Trenton Garner on "Disentangling the impacts of different stressors in a complex host pathogen system", **Distinction (74%)**
4. 2016 - 19, **Mumbai University, St. Xavier's College, India**
Integrated BSc, majoring in Zoology and Biochemistry, **Distinction (78%)**
 - Courses - Evolutionary biology, Ethology, Developmental Biology, Entomology, Biostatistics, Biochemical Pathways
5. 2015, **Delhi Public School, Navi Mumbai, India**
 - Subjects - Biology (96%), Physics (95%), Chemistry (92%), Psychology (83%), English (91%)

Professional Appointments

1. 2023 - 26, **Biology Graduate Student Association, University of Cincinnati, USA**
Faculty Representative
2. 2022, **Biology Graduate Student Association, University of Cincinnati, USA**
Graduate Student Government Representative
3. June - August 2019, **Bombay Natural History Society, India**
Intern - Wetlands Programme. Organisation and digitisation of long-term bird ringing data (Supervisor: Tuhina Katti)
4. September 2018 - September 19, **Nature Conservation Foundation, Mumbai, India**
Part-time volunteer. Collected and analysed data on shark landing and bycatch from fisheries for conservation planning
5. April - May 2018, **Universidad San Francisco de Quito, Ecuador**.
Volunteer - DNA extraction and PCR, bird captures using mist nets (Supervisor: Dr Elisa Bonaccorso)
6. December 2018, 2017, **Green Line Environmental NGO, Mumbai, India**
Volunteer - Conducting bird censuses for a biodiversity assay of Aarey, Mumbai, India
7. 2018, **Zoology board of Studies, Mumbai University, St Xavier's College, India**.

Student member - suggesting revisions to the second year BSc Zoology syllabus

Publications

1. Kelley M., Rathore S., Chandrasegaran K., Herbert C., Sabile C.E.G., Wood T., Joves J., Palacios A., Hamal B., Tompkin J., Susanto E., Uhran M., Ledezma-Ramírez A., Chen S., Singh K., Rust R., Solorio M.M., Khalid M.S., **Mitra A.T.**, Vinauger C., Buschbeck E.K., Limbach P.A., Benoit J.B. (2026) Microbiome derived queuine vitamers underlie tyrosine metabolism and predator avoidance in mosquito larvae. *Commun. biol.* DOI: [10.1038/s42003-026-10537-w](https://doi.org/10.1038/s42003-026-10537-w)
2. **Mitra A.T.**, Rathore S., Jester A., Hyland-Brown R., Hassert J., Benoit J.B., Stowasser A., Buschbeck E.K. (2025) Cataract induction in an arthropod reveals how lens crystallins contribute to the formation of biological glass. *PLoS One* 20.6 e0325229. DOI: [10.1371/journal.pone.0325229](https://doi.org/10.1371/journal.pone.0325229)
3. Rathore S., **Mitra A.T.**, Hyland-Brown R., Jester A., Layne J.E., Benoit J.B., Buschbeck E.K. (2024) Osmosis as nature's method for establishing optical alignment. *Curr. biol.* 34.7: 1569-1575. DOI: [10.1016/j.cub.2024.02.052](https://doi.org/10.1016/j.cub.2024.02.052)
4. **Mitra A.T.**, Womack M.C., Gower D.J., Clark B., Streicher J.W., Bell R.C., Schott R.K., Fujita M.K., Thomas K.N. (2022) Ocular lens morphology is influenced by ecology in frogs and toads. *Proc. R. Soc. B.* 289: 20220767. DOI: [10.1098/rspb.2022.0767](https://doi.org/10.1098/rspb.2022.0767)
5. Rosa G.M., Botto G.A., **Mitra A.T.**, de Almeida J.S., Hofmann M., Leung W.T., Alves de Matos A.P., Caeiro M.F., Froufe E., Loureiro A., Price S.J., Owen C., Rebelo R., Soares C. (2022). Invasive fish disrupt host-pathogen dynamics leading to amphibian declines. *Biol. Conserv.* 276, 109785. DOI: [10.1016/j.biocon.2022.109785](https://doi.org/10.1016/j.biocon.2022.109785)

Awards and Fellowships

1. 2024 - Present, **Graduate Assistant Scholarship, University Graduate Assistantship**, University of Cincinnati (\$27,775 annually)
2. 2026 - **SDB Trainee Travel Assistance**, Society for Developmental Biology (\$300)
3. 2026 - **Dr William Spoor Scholarship**, University of Cincinnati (\$1500)
4. 2026 - **Research Discovery & Endowment Fund**, University of Cincinnati (\$500)
5. 2025 - **Hibbitt Endowed Education Fund**, Marine Biological Laboratory (\$7,200)
6. 2024, **Graduate Student Government Graduate Research Fellowship Award**, University of Cincinnati (\$1,700)
7. 2023 - 24 - **IRiS Graduate Research Fellowship**, University of Cincinnati (\$3,500)
8. 2023 - **Carl Gans Travelling Fellowship**, Society for the Study of Amphibians and Reptiles (\$800)
9. 2022 - 26 - **Weiman/Benedict Awards**, University of Cincinnati (\$3,866)
10. 2019 - 20, **Ravi Sankaran Fellowship**, Inlaks Shivdasani Foundation (\$53,366)

Conferences and Presentations

1. July 2026, From compound to camera: developmental patterning of eyes in larval insects. **Society for Developmental Biology 85th Annual Meeting** (Talk)
2. June 2026, **Invited speaker**: From compound to camera: developmental patterning of eyes in larval insects. **St. Xavier's College, Mumbai, India** (Seminar)
3. June 2026, From compound to camera: developmental patterning of larval eyes in insects. **Novelty, Co-option & Divergence During Gene Network Evolution** Workshop, East Sussex, UK. (Talk)
4. October 2025, How lens crystallins form biological glass, **Institute for Research in Sensing (IRiS) Ignite** Biennial Meeting. (Talk)
5. September 2025, Establishing molecular parallels between eye types, **Society for**

Developmental Biology Midwest Regional Meeting. (*Poster,Flash Talk*)

6. January 2025, Investigating the role of the transcription factor Cut in the development of insect camera-type eyes, **Society for Integrative and Comparative Biology (SICB) Annual Meeting**. (*Talk*)
7. January 2024, Deconstructing a complex lens, **Society for Integrative and Comparative Biology (SICB) Annual Meeting**. (*Talk*)
8. July 2023, Deconstructing a complex lens, **International Conference on Invertebrate Vision (ICIV)** Bäckaskog Castle, Sweden (*Poster*)
9. July 2023, **Invited speaker**: Ocular lens morphology is influenced by ecology in frogs and toads, **Joint Meeting of Ichthyology and Herpetology (JMIH; Amphibian Vision Symposium)**, Norfolk, Virginia, USA. (*Talk*)
10. November 2022, Exploring the architecture of a complex lens, **Sensorium**, (Regional conference on sensory biology), Cincinnati, Ohio, USA. (*Poster*)
11. January 2021, Lens morphology is influenced by ecology in frogs and toads (Amphibia: Anura) **Society for Integrative and Comparative Biology (SICB) Virtual Meeting**. (*Talk*)
12. March 2020, The Evolution of Lenses in Frogs and Toads **Natural History Museum**, London, UK. (*Talk*)

Teaching

1. 2026, **Undergraduate research mentor** to Stephen Worthington, **University of Cincinnati**, USA
2. 2026, **Masters student mentor** to Maryia Sikirzhitskaya, **University of Cincinnati**, USA
3. 2026, **Prospective rotating PhD student mentor** to Kaladin White, **University of Cincinnati**, USA
4. 2025 - 26, **Undergraduate research mentor** to Claire Weberding, **University of Cincinnati**, USA
5. 2025, **Undergraduate research mentor** to Caden Pollock, **University of Cincinnati**, USA
6. 2024, **Undergraduate research mentor** to Augusta Jester, **University of Cincinnati**, USA
7. 2024, Teaching Assistant, **Cell Biology Laboratory**, **University of Cincinnati**, USA
8. 2023 - 25, Teaching Assistant, **Neurophysiology Laboratory**, **University of Cincinnati**, USA
9. 2022, 24, Teaching Assistant, **Introductory Biology Laboratory**, **University of Cincinnati**, USA

Science Communication & Outreach

1. 2024, **volunteer** for **Biology Day**; outreach event conducting visual ecology experiments with high-school students. University of Cincinnati, Ohio, USA
2. 2023, participant in the Annual Institute of Research in Sensing (IRiS) **Light Working Group** on the importance of light for humankind and the planet. To be developed into a book (In preparation). University of Cincinnati, Ohio, USA
3. 2023, **volunteer** for **Earth Day outreach**; public engagement event on arthropod biology with an interactive live-insect booth. Rhinegeist Brewery, Cincinnati, Ohio, USA
4. 2019, wrote on brood parasites for **The Quagga**, science blog of the SciComm students of University College London, the Natural History Museum and the Zoological Society of London
5. 2019, wrote on Lepidodendrales for Volume 1, **Acta Botanica**, annual departmental magazine of the botany department. St. Xavier's College, Mumbai, India
6. 2018 **content editor, designer** and 2019 **Student editor-in-chief** for **Imprint**, annual magazine of the department of zoology. St. Xavier's College, Mumbai, India
7. 2018, wrote on raft building in fire-ants for **Lignum Vitae**, annual departmental magazine of

- the life sciences department. St. Xavier's College, Mumbai, India
8. 2017, **graphic designer** for *Paradigm*, the St. Xavier's College Science festival

Research Skills:

Statistics & Programming

- Foundational statistics, linear & non-linear modelling and model selection methods
- Phylogenetic comparative methods in R (MSc level course taught by Dr Natalie Cooper)
- R and R Studio for statistical computing (Course taught by Dr Natalie Cooper, NHM London)
- Git and GitHub (Course taught by Dr Natalie Cooper, NHM, London)
- Basic Python for programming
- Basic MATLAB for programming

Imaging

- Micro-CT (scanning, staining and specimen preparation)
- Avizo software for 3D visualisation and analysis
- Transmission electron microscopy (imaging, staining and sample preparation)
- Scanning electron microscopy (sample preparation, imaging)
- Confocal, light sheet microscopy (imaging, sample preparation, cryosectioning and analysis)
- Micro-ophthalmoscopy

Molecular techniques

- RNA interference in arthropods
- Quantitative PCR
- Immunohistochemistry
- *In-situ* hybridisation
- Cryosectioning
- Micro-injection

Physiology

- Electroretinography (ERG)
- Intracellular electrophysiology

Insect rearing

- Coleoptera (Sunburst diving beetle, *Thermonectus marmoratus*)
- Hymenoptera (Red-headed pine sawfly, *Neodiprion lecontei*; Northern paper wasp, *Polistes fuscatus*)
- Strepsiptera (Twisted-wing insect, *Xenos peckii*)
- Mantodea (Chinese mantis, *Tenodera sinensis*)
- Odonata (Blue riverdamsel, *Pseudagrion microcephalum*)
- Blattodea (Madagascar hissing cockroach, *Gromphadorhina portentosa*)
- Diptera (Fruit fly, *Drosophila melanogaster*, Blow fly, *Calliphora*)
- Lepidoptera (Papilionidae, Sphingidae, Lycaenidae, Nymphalidae)

Field

- Transect sampling, bird capture via mist-netting for blood sampling, morphometric measurements
- Taxonomy and identification of birds (Indian subcontinent), sharks (Arabian Sea)

Population Modelling

- SCTI Tools (Vortex, Outbreak and MetaModel Manager softwares for population viability analysis, species risk assesment and metamodel analysis)

Scientific illustration and art

- Traditional media (watercolour, acrylic, pencil, marker, sculpture),
- Adobe, Affinity photoshop, indesign and illustrator for graphic design
- Skeleton assembly and taxidermy (course taught by Dr Santosh Gaikwad)
- Photography, videography

Languages:

Bangla, English, Marathi, Hindi, basic Spanish

References:

1. **Professor Elke K. Buschbeck**

Rieveschl Hall 800Q
Department of Biological Sciences
University of Cincinnati

2. **Professor Joshua B. Benoit**

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Department of Biological Sciences
University of Cincinnati